

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

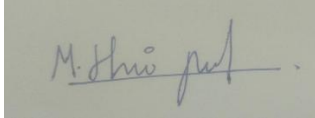
Total Marks: 30

Code of Conduct

I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

1. Understanding of the Case

The retail chain has implemented a cloud-based inventory management system that connects multiple retail stores through cloud services. Point-of-Sale (POS) systems and management dashboards act as clients that communicate with centralized cloud services using APIs. The cloud stores inventory data, transaction records, and analytics information, enabling real-time inventory monitoring and demand forecasting.

The objective is to improve inventory management, reduce stock shortages, optimize product availability, and support business decisions through cloud-based analytics.

2. Technical and Business Requirements

Technical Requirements

- Connect multiple store devices to a centralized cloud.
- Provide secure communication using APIs.
- Store inventory, sales, and customer transaction data.
- Analyze sales trends for demand forecasting.
- Ensure secure access through VPN, firewall, and API Gateway.
- Provide high availability and scalability.

Business Requirements

- Maintain accurate inventory across all stores.
- Reduce overstocking and stock shortages.
- Improve customer satisfaction through product availability.
- Reduce operational costs.
- Support business growth without replacing infrastructure.

3. Cloud Computing Concepts Applied

Infrastructure as a Service (IaaS)

Virtual machines, networking, and storage provide scalable infrastructure for hosting inventory services.

Platform as a Service (PaaS)

Cloud analytics services process sales data and perform demand forecasting without managing servers.

Software as a Service (SaaS)

Managers access dashboards through web applications to monitor inventory and sales.

Cloud Storage

Stores:

- Product database
- Sales transactions
- Customer purchase history
- Analytics datasets

Web APIs

REST APIs enable communication between:

- POS systems
- Mobile applications
- Management dashboards
- Cloud database

4. Cloud Architecture Analysis

The architecture consists of several interconnected components:

- POS terminals collect customer purchase information.
- API Gateway securely receives requests.
- Cloud services process inventory updates.
- Cloud storage maintains centralized inventory records.
- Analytics platform predicts future demand.

- Dashboards display reports to managers.

This architecture enables every store to access the same updated inventory information in real time.

5. Security Mechanisms

The system protects business data using multiple security layers.

VPN

Provides secure communication between stores and cloud infrastructure.

Firewall

Blocks unauthorized traffic entering the network.

API Gateway

Authenticates API requests, controls traffic, and prevents misuse.

Access Control

Employees receive permissions according to their job responsibilities.

Data Encryption

Sensitive business information remains protected during storage and transmission.

6. Solution Design

A practical cloud solution includes:

- POS devices send sales information through REST APIs.
- API Gateway authenticates requests.
- Cloud application updates inventory instantly.
- Cloud database stores inventory records.
- Analytics platform forecasts future demand.
- Dashboard displays inventory reports for managers.
- Backup and disaster recovery protect business continuity.

7. Benefits

- Real-time inventory tracking
- Centralized management
- High scalability
- Faster business decisions
- Improved demand forecasting
- Reduced inventory cost
- Better customer satisfaction
- High availability

2.Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Case Study 2: Cloud-Based Document Collaboration System

1. Understanding of the Case

The enterprise has developed a cloud-based document collaboration platform similar to Google Docs. Users can create, edit, and share documents simultaneously through web browsers. Documents are stored in distributed cloud storage, while APIs synchronize changes among multiple users in real time.

The primary objective is to provide secure, highly available, and globally accessible document collaboration.

2. Technical and Business Requirements

Technical Requirements

- Real-time collaborative editing
- Distributed cloud storage
- Version control
- Low latency worldwide
- Web API synchronization
- High availability
- Strong security

Business Requirements

- Improve employee collaboration
- Enable remote work
- Prevent document loss
- Increase productivity
- Ensure secure sharing

3. Cloud Computing Concepts Applied

SaaS

Users access the application directly through web browsers without software installation.

PaaS

Cloud services manage synchronization, document processing, and version control.

IaaS

Virtual servers, storage, and networking provide scalable infrastructure.

Distributed Cloud Storage

Documents are replicated across multiple regions for availability and reliability.

REST APIs

Enable synchronization between:

- Browsers
- Mobile devices
- Cloud storage
- Collaboration services

4. Architecture Analysis

The workflow operates as follows:

1. User edits a document.
2. Browser sends changes through APIs.
3. Collaboration server validates updates.
4. Version control stores changes.
5. Cloud storage saves new versions.
6. Other connected users instantly receive updates.

This architecture enables simultaneous editing without data conflicts.

5. Security Mechanisms

Identity and Access Management (IAM)

Only authenticated users can access documents.

Encryption

Protects documents during storage and transmission.

Role-Based Access Control

Permissions include:

- Owner
- Editor
- Viewer

Version Control

Maintains document history for recovery.

Backup

Prevents permanent data loss.

6. Solution Design

A complete solution includes:

- Browser clients
- API Gateway
- Collaboration engine
- Version control service
- Distributed cloud storage
- Authentication server
- Monitoring system

This architecture ensures secure, reliable, and scalable collaboration.

7. Benefits

- Real-time collaboration
- Global accessibility
- Automatic backup
- High availability
- Secure document sharing
- Easy scalability
- Reduced maintenance cost
- Better productivity

3.Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time

1. Understanding of the Case

A startup has developed a cloud-based food ordering application that allows customers to order food using web browsers and mobile applications. Frontend clients communicate with backend cloud services using REST APIs. Cloud storage stores menus, invoices, and customer data, while Kubernetes and virtual machines provide scalable infrastructure.

The goal is to provide reliable, secure, and scalable online food ordering services.

2. Technical and Business Requirements

Technical Requirements

- Mobile and web clients
- RESTful APIs
- Cloud storage
- Kubernetes
- Virtual machines
- Authentication
- HTTPS communication
- Role-based access control

Business Requirements

- Fast order processing
- Handle peak customer traffic
- Secure customer information
- High service availability
- Easy scalability

3. Cloud Computing Concepts Applied

Infrastructure as a Service (IaaS)

Virtual machines host backend applications.

Platform as a Service (PaaS)

Managed Kubernetes automatically deploys and scales application containers.

SaaS

Customers access food ordering services through browsers and mobile applications.

Cloud Storage

Stores:

- Menu images
- Customer information
- Order history
- Payment invoices

REST APIs

Connect:

- Mobile applications
- Web browsers
- Payment systems
- Restaurant dashboards

4. Architecture Analysis

The architecture works as follows:

1. Customer places an order.
2. Frontend sends request through REST API.
3. API validates authentication token.
4. Backend processes order.

5. Cloud database stores transaction.
6. Restaurant receives notification.
7. Customer receives order confirmation.

Kubernetes automatically creates additional containers when user traffic increases.

5. Security Mechanisms

HTTPS

Encrypts communication between client and server.

Authentication Tokens

Verify user identity before processing requests.

Role-Based Access Control (RBAC)

Different permissions for:

- Customers
- Restaurant staff
- Administrators

Data Encryption

Protects sensitive customer information.

Secure Cloud Storage

Safeguards invoices and menu images.

6. Solution Design

An efficient solution includes:

- Web and mobile frontend
- API Gateway
- Authentication service
- Kubernetes cluster
- Backend microservices

- Cloud database
- Object storage
- Monitoring and logging
- Backup and disaster recovery

7. Benefits

- High scalability
- Fast response time
- Secure transactions
- Easy deployment
- Automatic scaling
- High availability
- Better customer experience
- Reduced operational cost